Notes: Bresnahan & Reiss (JPE, 1991)
Entry and Competition in Concentrated Markets

Contents

1 Big picture

- **Question.** How much does actual entry change competitive conduct in concentrated local markets when researchers do not observe firm-level prices, quantities, or marginal costs?
- **Core idea.** If a local market needs a much larger population to support the second firm than the first firm, the first entrant must be earning high margins or entrants must face higher costs. Entry-threshold ratios therefore reveal information about competition even when direct margin data are unavailable.
- **Empirical setting.** The paper studies 202 geographically isolated local markets, mostly county-seat towns in the western United States, and counts the number of doctors, dentists, druggists, plumbers, and tire dealers.
- **Main empirical object.** The demand entry threshold S_N is the market size required to support N firms at zero profits. Ratios such as S_2/S_1, S_3/S_2, and S_5/S_N summarize how quickly competition intensifies after entry.
- **Core result.** Most of the competitive change occurs with the second and third firms. Once a market has roughly three to five firms, additional entry has little effect on conduct in most of the studied industries.
- **Validation.** Tire-price evidence supports the entry-threshold results: prices fall when markets move from one or two dealers to three or more, but prices then level off within the small-market oligopoly range.
- **Economic takeaway.** A small number of actual competitors can discipline local-market conduct substantially, but potential competition alone is not enough; the jump from monopoly/duopoly to triopoly is especially important.

2 Incremental contribution of the paper

- **Entry thresholds as an empirical sufficient statistic.** The paper turns the minimum market size needed to support N firms into an empirical measure of how entry changes conduct. This lets researchers infer competition without observing prices or costs.
- **Beyond simple monopoly-entry models.** Earlier work estimated first and second entry thresholds. This paper extends the method to third, fourth, and fifth entrants and allows U-shaped average costs and entrant cost differences.
- **Scale-free conduct measures.** The paper emphasizes ratios such as S_{N+1}/S_N rather than levels. Ratios are easier to compare across industries because they remove units of market size and focus on incremental entry effects.
- **Ordered discrete-choice structure for market structure.** The number of firms in a market is treated as the observable outcome of latent profit inequalities. This creates an ordered-probit style model of equilibrium firm counts.
- **Credible local-market design.** The paper uses isolated towns so that local population is a meaningful proxy for market size and so that the relevant set of competitors is easier to define.
- **Cross-validation with prices.** The paper separately collects tire-price quotes to test whether the threshold-based conduct inference agrees with actual price patterns. This is important because the main entry-threshold method is otherwise indirect.
- **Industrial-organization impact.** The paper helped establish entry models as a way to study competition in local markets and influenced later empirical IO work on entry, market structure, and competition when price data are limited.

3 Data and measurement

3.1 Market definition and sample construction

- The sample contains 202 isolated local markets, typically county seats in the western United States.
- Towns are selected to be far from other towns and large metropolitan areas, reducing contamination from neighboring markets.
- The main demand proxy is central town population, supplemented by nearby population, population growth, and commuting measures.
- Firm counts come from business listings, phone books, trade sources, and direct verification for some markets.

Interpretation: The sample design tries to make each town a self-contained market. This is crucial because the entry-threshold approach relies on linking local population to the number of firms supported locally.

3.2 Industries studied

- The paper focuses on doctors, dentists, druggists, plumbers, and tire dealers.
- The authors avoid broad retail categories, such as grocery or clothing stores, because those categories sell many products and make entry counts harder to interpret.
- The final industries have enough observations across firm-count categories to estimate thresholds for multiple entrants.

Interpretation: These industries are services or local retail categories where local population should be a strong determinant of demand and where market boundaries are more plausible.

3.3 Market size variables

The market-size index is normalized in units of current town population:

$$S(Y, \lambda) = TPOP + \lambda_1 OPOP + \lambda_2 PGRW + \lambda_3 NGRW + \lambda_4 OCTY.$$

- $TPOP$ is town population.
- $OPOP$ is nearby population.
- $PGRW$ and $NGRW$ measure positive and negative population growth.
- $OCTY$ measures commuters out of the county, capturing possible demand leakage to outside markets.

Interpretation: The market-size index turns demographic information into an effective local demand measure. If nearby residents shop in the town, $OPOP$ raises market size; if residents commute out and shop elsewhere, $OCTY$ can lower effective market size.

3.4 Profit shifters and demographic controls

- Demand/profit shifters include birth rate, elderly population share, per capita income, heating degree days, housing units, fraction of land in farms, land values, and housing values.

- Different industries use different subsets of these variables depending on relevance.
- The model allows industry-specific profit shifters because, for example, elderly population may matter more for doctors, and housing variables may matter more for plumbers.

Interpretation: The controls are not the main object; they improve threshold estimates by accounting for demand and cost differences across towns that are not captured by raw population alone.

4 Model and empirical specifications

4.1 Demand and profits

The paper starts with market demand

$$Q^T = d(Z, P)S(Y),$$

where $d(Z, P)$ is representative-consumer demand and $S(Y)$ is the number of consumers.

For N firms, per capita variable profit is denoted V_N. The zero-profit condition for N firms is

$$\Pi_N = V_N S(Y) - F_N - B_N = 0,$$

where F_N represents fixed costs and B_N captures entry-related barriers or cost differences.

Interpretation: Entry thresholds arise from this zero-profit condition. A market supports N firms when variable profits times market size cover the relevant fixed and entry costs.

4.2 Entry thresholds

The entry threshold S_N is the market size required to support N firms:

$$S_N = \frac{F_N + B_N}{V_N}.$$

The key empirical statistic is a ratio:

$$\frac{S_{N+1}}{S_N}.$$

Interpretation: If entry sharply reduces profits per firm, S_{N+1}/S_N will be much larger than one. If additional entry has little effect on conduct, the ratio should approach one.

4.3 Variable-profit specification

The empirical model writes per capita variable profits as

$$V_N = \alpha_1 + X\beta - \sum_{n=2}^{N} \alpha_n.$$

- $\alpha_1 + X\beta$ is monopoly per capita variable profit.
- α_n measures the incremental reduction in per capita profits when the nth firm is active.
- Fixed costs are modeled analogously with a baseline and incremental terms.

Interpretation: The structural shifts in V_N are the way the model translates changes in firm counts into changes in competition.

4.4 Ordered-probit market-structure likelihood

Let Π_N be the latent profit from operating with N firms. The probabilities of observing market structures are based on threshold inequalities:

$$Pr(N = 0) = Pr(\Pi_1 < 0),$$
$$Pr(N = k) = Pr(\Pi_k \geq 0,\ \Pi_{k+1} < 0), \quad k = 1, 2, 3, 4,$$
$$Pr(N \geq 5) = Pr(\Pi_5 \geq 0).$$

Interpretation: The dependent variable is the observed number of firms, not price. The ordered-probit structure links observed market structure to latent profitability conditions.

5 Identification and empirical design

5.1 What identifies entry thresholds

- Cross-market variation in population and demographics generates variation in market size.
- Cross-market variation in the number of incumbent firms maps that market size into entry thresholds.
- Geographic isolation helps ensure that firm counts are local market counts and that town population is relevant demand.

Interpretation: The approach is a cross-sectional comparative-statics design: larger isolated markets reveal how many firms can be supported at different demand levels.

5.2 Why the design is informative about competition

- If the second entrant strongly lowers margins, a much larger market is needed to support two firms than one firm.
- If the third entrant adds little competitive pressure, the market-size increment needed for three firms relative to two should be smaller.
- Ratios close to one imply that additional firms require roughly proportional demand, suggesting that conduct has already approached the relevant local competitive norm.

Interpretation: The identification is indirect: entry thresholds do not literally observe markups. They infer conduct from how market size must expand to accommodate more firms.

5.3 Threats and how the paper addresses them

- **Overlapping markets.** Isolated-town selection and alternative market-definition tests address whether consumers or firms cross market boundaries.
- **Different entrant costs.** Tests for equal fixed costs and robustness specifications check whether high threshold ratios reflect entry barriers rather than conduct.
- **Demand heterogeneity.** Demographic controls and industry-specific variables account for nonpopulation demand differences.

- **Price validation.** Tire price regressions test whether threshold-based inferences align with observed prices.

Interpretation: The paper cannot fully separate all cost and conduct channels, but it makes the maintained assumptions explicit and checks the main interpretation with price data.

6 Section-by-section study guide

- **Introduction.** Motivates the need for a method that can distinguish between contestable-market views and actual-entry views when price/cost data are unavailable.
- **Entry and market size.** Develops entry thresholds and explains why ratios of thresholds measure the competitive impact of entry.
- **Retail and professional market entry thresholds.** Describes the isolated-market sample, industries, market-size variables, and ordered-probit estimation.
- **Supplemental tire price evidence.** Uses actual tire-price quotes to validate the threshold-based inference.
- **Conclusion.** Emphasizes that most competitive change occurs with the second and third entrants, while later entrants in small markets have limited additional effects.

7 Tables and figures

7.1 Figure 1: Breakeven Firm Demand and Margins + Table 1: Successive Entry Threshold Ratios for a Cournot Oligopoly Model with Linear Demand and Constant Marginal Costs

Read:

- Figure 1 explains the logic of entry thresholds visually. A monopolist breaks even at a low demand level but earns a positive price-cost margin.
- As market demand expands, the demand curve rotates outward and additional entrants can be supported.
- Table 1 shows that in a Cournot example, both margins and threshold ratios fall as N rises.
- With constant marginal costs, $S_2/S_1 = 2.25$, $S_3/S_2 = 1.78$, and $S_4/S_3 = 1.56$, showing large early entry effects.
- By $N = 20$, ratios are near 1.10 and margins are much smaller, showing convergence toward competitive conduct.

Detailed interpretation: Figure 1 makes the key identification move transparent: margins are not observed, so market size at the entry boundary substitutes for direct markup data. Table 1 then gives a numerical benchmark for interpreting observed threshold ratios. Large early ratios are consistent with monopoly/duopoly market power; ratios near one are consistent with little incremental effect of additional entry.

Economic message: The theory predicts that entry matters most early. The empirical question is whether the real markets display the same rapid decline in threshold ratios.

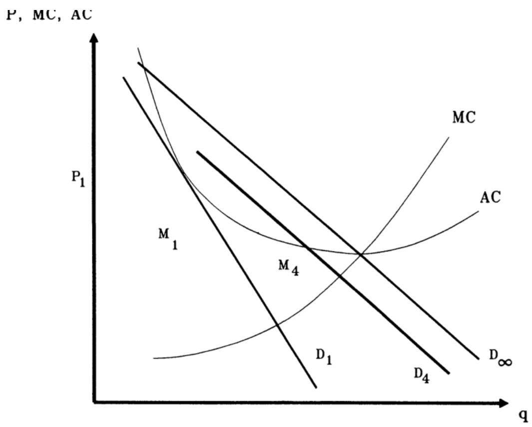

FIG. 1.—Breakeven firm demand and margins

Figure 1: Breakeven firm demand and margins

TABLE 1

SUCCESSIVE ENTRY THRESHOLD RATIOS FOR A COURNOT OLIGOPOLY MODEL WITH LINEAR DEMAND AND CONSTANT MARGINAL COSTS

Number of Firms	k = 0, MES = ∞ s_{N+1}/s_N (1)	$P_N - MC_N$ (2)	k = 2, MES = 1.58 s_{N+1}/s_N (3)	$P_N - MC_N$ (4)	k = 10, MES = .71 s_{N+1}/s_N (5)	$P_N - MC_N$ (6)
1	2.25	7.5	2.17	6.3	2.01	.8
2	1.78	5.0	1.64	3.8	1.52	.4
3	1.56	3.8	1.42	2.7	1.34	.3
4	1.44	3.0	1.31	2.1	1.25	.2
5	1.36	2.5	1.24	1.7	1.20	.2
20	1.10	.8	1.06	.4	1.05	.0
∞	1.00	.0	1.00	.0	1.00	.0

NOTE.—Price minus marginal cost equals $15 - (Q/S) - 2kq$. Fixed costs equal five. MES denotes minimum efficient scale

Table 1: Successive entry threshold ratios for Cournot oligopoly

7.2 Figure 2: Number of Towns by Town Population + Table 2: Market Counts by Industry and Number of Incumbents

Read:

- Figure 2 shows that the sample contains many small towns but also a tail of larger local markets.
- This variation is essential because the model needs markets of different sizes to infer the population required to support each number of firms.
- Table 2 shows the distribution of firm counts across industries. The final five industries have observations across low and high firm-count categories.
- The broad distribution of counts makes it possible to estimate entry thresholds for the second through fifth firms.

Detailed interpretation: Figure 2 is a data-support figure rather than a result figure. It shows that the sample is not concentrated at one market size. Table 2 provides the discrete-choice dependent variable: the number of firms by industry and market.

Economic message: The sample design provides the empirical variation needed for entry-threshold estimation.

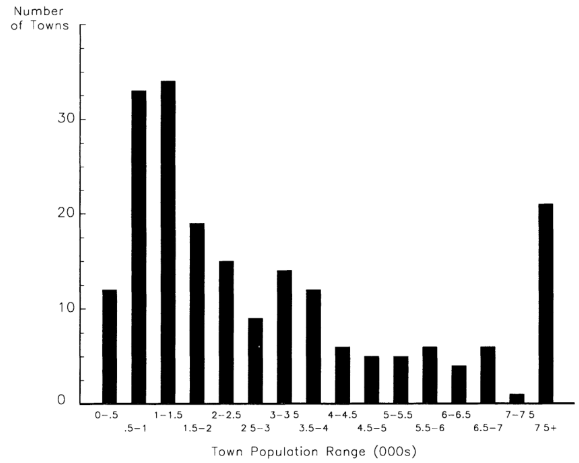

Figure 2: Number of towns by town population

TABLE 2

Market Counts by Industry and Number of Incumbents

STRY	Number of Firms $N = 0$	$N = 1$	$N = 2$	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N \geq$
gists	28	62	68	23	8	6	3	4
ors	37	61	36	16	11	7	6	28
ists	32	67	39	15	12	12	4	21
bers	71	47	26	21	10	4	6	17
dealers	45	39	39	24	13	15	6	21
ers	95	66	23	9	3	6	0	0
cians	173	19	5	1	4	0	0	0
ticians	10	26	19	24	26	19	11	67
metrists	68	85	36	7	3	3	0	0
ricians	60	54	32	17	10	5	7	17
rinarians	53	80	41	21	5	0	1	1
e theaters	90	72	25	10	5	0	0	0
mobile dealers	38	44	54	35	25	2	1	3
ing contractors	117	40	19	8	4	8	3	3
ing contractors	153	30	13	5	1	0	0	0
equipment dealers	90	39	23	19	17	9	1	4

RCE.—Authors' tabulations from American Business Lists, Inc.

Table 2: Market counts by industry and number of incumbents

7.3 Figure 3: Dentists by Town Population + Table 3: Sample Market Descriptive Statistics

Read:

- Figure 3 shows how the distribution of dentist counts shifts with town population.
- Very small towns often have zero or one dentist; larger towns increasingly have three or more.
- Table 3 reports the main variables used in the entry model: firm counts, population variables, and demographic/profit shifters.
- The average town population is 3.74 thousand, while firm counts vary substantially across industries.

Detailed interpretation: Figure 3 gives a concrete example of the ordered-response structure: as market size increases, the probability of higher firm counts rises. Table 3 shows that the model does not rely only on raw town population; it adjusts for nearby population, growth, commuting, income, age structure, climate, housing, and land-use characteristics.

Economic message: Firm counts rise with market size, but the rate at which they rise reveals how quickly entry erodes profits.

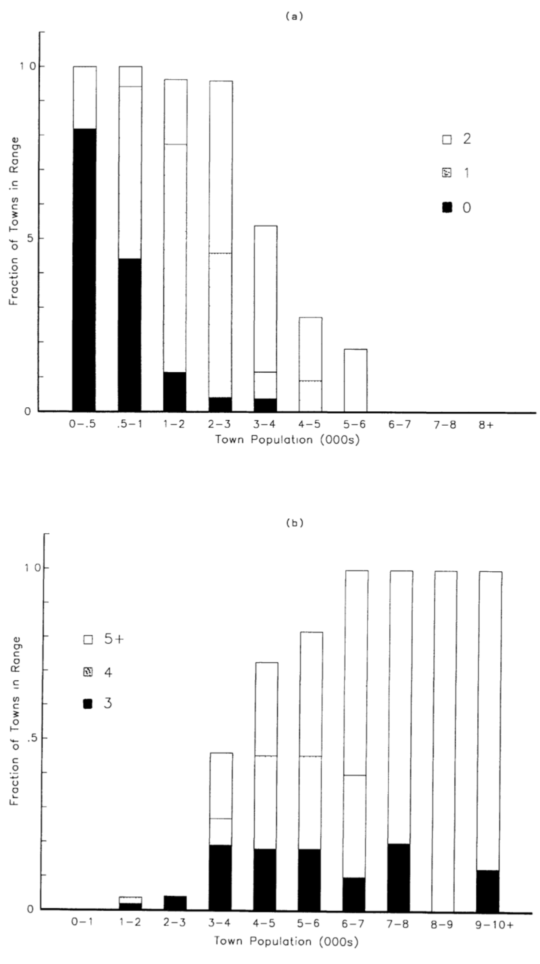

Figure 3: Dentists by town population

TABLE 3

SAMPLE MARKET DESCRIPTIVE STATISTICS

Variable	Name	Mean	Standard Deviation	Min	Max
Firm counts:					
Doctors	DOCS	3.4	5.4	.0	45.0
Dentists	DENTS	2.6	3.1	.0	17.0
Druggists	DRUG	1.9	1.5	.0	11.0
Plumbers	PLUM	2.2	3.3	.0	25.0
Tire dealers	TIRE	2.6	2.6	.0	13.0
Population variables (in thousands):					
Town population	TPOP	3.74	5.35	.12	45.09
Negative TPOP growth	NGRW	−.06	.14	−1.34	.00
Positive TPOP growth	PGRW	.49	1.05	.00	7.23
Commuters out of the county	OCTY	.32	.69	.00	8.39
Nearby population	OPOP	.41	.74	.01	5.84
Demographic variables:					
Birth ÷ county population	BIRTHS	.02	.01	.01	.04
65 years and older ÷ county population	ELD	.13	.05	.03	.30
Per capita income ($1,000's)	PINC	5.91	1.13	3.16	10.50
Log of heating degree days	LNHDD	8.59	.47	6.83	9.20
Housing units ÷ county population	HUNIT	.46	.11	.29	1.40
Fraction of land in farms	FFRAC	.67	.35	.00	1.27
Value per acre of farmland and buildings ($1,000's)	LANDV	.30	.23	.07	1.64
Median value of owner-occupied houses ($1,000's)	HVAL	32.91	14.29	9.90	106.0

Table 3: Sample market descriptive statistics

7.4 Table 4: Baseline Specifications + Table 5: Entry Threshold Estimates

Read:

- Table 4 reports ordered-probit profit specifications for the five industries.
- It includes market-size variables, demographic profit shifters, variable-profit shifts by number of firms, fixed-cost terms, and log likelihoods.
- Table 5 reports the core estimated entry thresholds and threshold ratios.
- The first threshold differs sharply by industry: doctors and dentists require roughly 700–900 people, druggists and tire dealers roughly 500, and plumbers about 1,400.
- The most important ratios are s_2/s_1 and s_3/s_2. They are often large for early entrants and much closer to one after the third entrant.

Detailed interpretation: Table 4 is the parameter backbone, while Table 5 is the economic translation. The paper's main claim comes from Table 5: entry-threshold ratios fall quickly with the number of firms. The table also shows heterogeneity across industries, with druggists displaying especially high early ratios and plumbers displaying smaller early ratios.

Economic message: Competition intensifies most strongly with early entry. Once three to five firms are present, additional entry often has small incremental effects.

TABLE 4

BASELINE SPECIFICATIONS

Variable Name	Doctors	Dentists	Druggists	Plumbers	Tire Dealers
OPOP (λ_1)	1.15	−.46	.08	.27	−.53
	(.85)	(.32)	(.37)	(.60)	(.43)
NGRW (λ_2)	−1.89	.63	−.30	.68	2.25
	(1.60)	(.85)	(.97)	(1.10)	(.75)
PGRW (λ_3)	1.92	−.35	−.24	−.45	.34
	(1.01)	(.41)	(.41)	(.36)	(.59)
OCTY (λ_4)	.80	2.72	.16	−.28	.23
	(1.26)	(.98)	(.34)	(.71)	(.94)
BIRTHS (β_1)	−.59	9.86	11.34		
	(6.57)	(8.29)	(10.10)		
ELD (β_2)	−.11	.22	2.61		−.49
	(.55)	(.74)	(.78)		(.75)
PINC (β_3)	−.00	.04	.02	.05	−.03
	(.00)	(.03)	(.02)	(.03)	(.04)
LNHDD (β_4)	.013	.28	.08	.003	.004
	(.05)	(.07)	(.06)	(.06)	(.06)
HUNIT (β_5)				.51	
				(.46)	
HVAL (β_6)				.42	
				(.03)	
FFRAC (β_7)					−.02
					(.08)
V_1 (α_1)	.63	−1.85	−.13	.06	.86
	(.46)	(.61)	(.58)	(.52)	(.45)
$V_1 - V_2$ (α_2)	.34		.29		.03
	(.17)		(.21)		(.15)
$V_2 - V_3$ (α_3)		.12	.19	.15	.15
		(.14)	(.17)	(.09)	(.10)
$V_3 - V_4$ (α_4)	.07	.20	.25	.07	
	(.05)	(.06)	(.14)	(.08)	
$V_4 - V_5$ (α_5)			.04	.04	.08
			(.12)	(.07)	(.05)
F_1 (γ_1)	.92	1.10	.91	1.28	.53
	(.30)	(.25)	(.29)	(.26)	(.23)
$F_2 - F_1$ (γ_2)	.65	1.84	1.34	1.04	.76
	(.30)	(.19)	(.35)	(.14)	(.21)
$F_3 - F_2$ (γ_3)	.84	1.14	1.77	.32	.46
	(.13)	(.46)	(.54)	(.28)	(.21)
$F_4 - F_3$ (γ_4)	.18		.06	.40	.60
	(.23)		(.70)	(.35)	(.12)
$F_5 - F_4$ (γ_5)	.42	.66	.51	.25	.12
	(.13)	(.60)	(.95)	(.35)	(.20)
LANDV (γ_L)	−1.02	−1.31	−.84	−1.18	−.74
	(.53)	(.37)	(.51)	(.48)	(.34)
Log likelihood	−233.49	−183.20	−195.16	−228.27	−263.09

NOTE.—Asymptotic standard errors are in parentheses

Table 4: Baseline specifications

TABLE 5

A. ENTRY THRESHOLD ESTIMATES

	ENTRY THRESHOLDS (000's)					PER FIRM ENTRY THRESHOLD RATIOS			
ESSION	S_1	S_2	S_3	S_4	S_5	s_2/s_1	s_3/s_2	s_4/s_3	s_5/s_4
ors	.88	3.49	5.78	7.72	9.14	1.98	1.10	1.00	.95
ists	.71	2.54	4.18	5.43	6.41	1.78	.79	.97	.94
gists	.53	2.12	5.04	7.67	9.39	1.99	1.58	1.14	.98
bers	1.43	3.02	4.53	6.20	7.47	1.06	1.00	1.02	.96
dealers	.49	1.78	3.41	4.74	6.10	1.81	1.28	1.04	1.03

B. LIKELIHOOD RATIO TESTS FOR THRESHOLD PROPORTIONALITY

ession	Test for $s_4 = s_5$	Test for $s_3 = s_4 = s_5$	Test for $s_2 = s_3 = s_4 = s_5$	Test for $s_1 = s_2 = s_3 = s_4 = s_5$
ors	1.12 (1)	6.20 (3)	8.33 (4)	45.06* (6)
ists	1.59 (1)	12.30* (2)	19.13* (4)	36.67* (5)
gists	.43 (2)	7.13 (4)	65.28* (6)	113.92* (8)
bers	1.99 (2)	4.01 (4)	12.07 (6)	15.62* (7)
dealers	3.59 (2)	4.24 (3)	14.52* (5)	20.89* (7)

E.—Estimates are based on the coefficient estimates in table 4. Numbers in parentheses in pt. B are degrees of freedom.
gnificant at the 5 percent level.

Table 5: Entry threshold estimates

7.5 Figure 4: Industry Ratios of s_5 to s_N by N + Table 6: Likelihood Ratio Tests for Equal Fixed Costs

Read:

- Figure 4 plots s_5/s_N for each industry and number of firms.
- The ratios are high at $N = 1$ and sometimes at $N = 2$, but they are close to one by $N = 3$ or $N = 4$.
- Table 6 tests whether fixed costs are equal across entrants.
- The equal-fixed-cost hypothesis is rejected for doctors, dentists, druggists, and tire dealers, but not for plumbers.
- Even when fixed costs differ, the constrained threshold ratios remain similar to the baseline ratios.

Detailed interpretation: Figure 4 is the clearest visual expression of the main result: most of the distance from monopoly-like conduct to the five-firm benchmark is eliminated after two or three active firms. Table 6 addresses an interpretation concern: high early threshold ratios may reflect higher entrant costs rather than competition. The paper acknowledges this issue and shows that the qualitative pattern survives the fixed-cost tests.

Economic message: The data point to a rapid early increase in competition, not a gradual linear decline in market power with each additional firm.

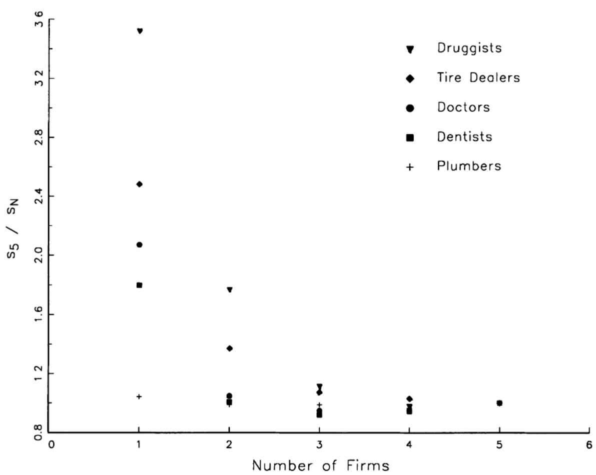

Figure 4: Industry ratios of s_5 to s_N by N

TABLE 6

LIKELIHOOD RATIO TESTS FOR EQUAL FIXED COSTS

Profession	Likelihood Value	Test Statistic	Degrees of Freedom	s_2/s_1	s_3/s_2	s_4/s_3	s_5/s_4
Doctors	247.80	28.63*	2	1.56	1.11	1.02	1.08
Dentists	187.84	9.29*	1	1.51	1.10	.98	1.01
Druggists	205.10	19.89*	4	1.68	1.55	1.16	1.09
Plumbers	231.69	6.84	3	.99	.99	1.07	1.08
Tire dealers	266.90	7.61*	2	1.32	1.24	1.07	1.11

* Significant at the 5 percent level.

Table 6: Likelihood ratio tests for equal fixed costs

7.6 Table 7: Likelihood Ratio Exclusion Tests for the Market Size Regressors + Table 8: Likelihood Ratio Exclusion Tests for the Variable Profits Regressors

Read:

- Table 7 asks whether the market-size index can omit nearby population, growth, and commuting variables.
- Several industries show meaningful changes in early ratios when these variables are omitted, supporting the richer market-size measure.
- Table 8 asks whether demographic profit shifters can be omitted from variable-profit equations.
- The druggist specification is especially sensitive to the omitted-variable test, while several other industries are less sensitive.

Detailed interpretation: These tables are robustness and specification-diagnostic tables. They show that the threshold estimates are not pure population-count exercises. Some controls matter because different industries depend on different demographic and local-economic conditions.

Economic message: Correctly measuring market size and local demand shifters matters for interpreting entry thresholds as competition measures.

TABLE 7

LIKELIHOOD RATIO EXCLUSION TESTS FOR THE MARKET SIZE REGRESSORS

Likelihood Value	Test Statistic	Variables Omitted	s_2/s_1	s_3/s_2	s_4/s_3	s_5/s_4
236.35	5.74	OPOP, OCTY	2.33	1.11	1.00	.98
183.84	1.30	PGRW, NGRW	1.78	1.11	.98	.98
195.61	.90	PGRW, OPOP, OCTY	2.08	1.581	1.16	1.03
228.74	.99	NGRW, OPOP, OCTY	1.05	.98	1.12	1.01
265.27	4.36	NGRW, OCTY	2.10	1.24	1.03	1.06

Table 7: Market-size regressor exclusion tests

TABLE 8

LIKELIHOOD RATIO EXCLUSION TESTS FOR THE VARIABLE PROFITS REGRESSORS

Likelihood Value	Test Statistic	Variables Omitted	s_2/s_1	s_3/s_2	s_4/s_3	s_5/s_4
233.52	.06	BIRTHS, ELD, PINC	1.98	1.10	1.00	.95
183.25	.11	ELD	1.76	1.09	.98	.94
228.29	.36	LNHDD	1.06	1.00	1.03	.96
263.92	1.66	ELD, PINC, LNHDD, LANDV	1.85	1.28	1.04	1.02

ggists specification did not have the absolute value of any coefficient smaller than its estimated standard error.

Table 8: Variable-profit regressor exclusion tests

7.7 Table 9: Entry Thresholds for Alternative Market Definitions + Table 10: Tire Price Sample Descriptive Statistics

Read:

- Table 9 tests weaker and stronger distance criteria for defining isolated markets, focusing on doctors and dentists.
- A stronger distance criterion produces larger early threshold ratios, consistent with cleaner market isolation.
- Table 10 describes the tire-price survey used to validate the threshold results.
- Mean and median tire prices are higher in concentrated local markets than in the urban comparison market.
- Monopoly and duopoly prices look similar, while prices in markets with three to five dealers are lower and more similar to each other.

Detailed interpretation: Table 9 checks a central design issue: if market boundaries are poorly defined, the estimated threshold ratios could be misleading. Table 10 introduces direct price data. Its descriptive statistics already suggest the same pattern as the threshold results: the big price change is not between every adjacent firm count but between very concentrated markets and somewhat less concentrated markets.

Economic message: Market definition matters, and direct tire-price data provide an external validity check for the threshold-based inference.

TABLE 9

ENTRY THRESHOLDS FOR ALTERNATIVE MARKET DEFINITIONS

PROFESSION	Weaker Distance Criterion s_2/s_1	Weaker Distance Criterion s_3/s_2	Weaker Distance Criterion s_4/s_3	Weaker Distance Criterion s_5/s_4	Stronger Distance Criterion s_2/s_1	Stronger Distance Criterion s_3/s_2	Stronger Distance Criterion s_4/s_3	Stronger Distance Criterion s_5/s_4
Dentists	1.13	.88	.94	.99	1.82	1.15	1.06	*
Doctors	1.05	1.07	1.10	1.01	1.93	1.02	1.01	*

* Not estimable because of small sample sizes.

original sample. The other half were not because they failed our market definition criteria. To limit data collection costs, we analyzed

Table 9: Entry thresholds for alternative market definitions

TABLE 10

TIRE PRICE SAMPLE DESCRIPTIVE STATISTICS

	Number of Tire Dealers in the Market: 1	2	3	4	5	1.5	Urban
Candidate phone listings	39	66	48	64	75	*	200+
Surveyed by us	36	22	19	28	21	20	19
At listed number	32	19	19	24	21	17	18
Would respond	28	19	19	23	20	14	17
Total prices quoted	76	52	50	64	49	36	62
Usable price quotations	42	31	40	57	45	17	59
	Sample Means						
Price	54.9	55.7	54.4	51.6	52.0	53.8	45.6
Tire mileage rating (000)	44.5	47.0	47.7	45.4	43.8	43.0	45.3
	Sample Medians						
Price	53.9	55.0	52.9	50.9	49.8	51.7	43.2
Tire mileage rating (000)	45	45	50	40	40	40	45

* Unknown.

Table 10: Tire price sample descriptive statistics

7.8 Table 11: Tire Price Regressions ($N = 282$)

Read:

- Table 11 estimates tire prices as a function of market structure, tire quality, wage costs, and brand effects.
- Urban markets have substantially lower prices than the isolated-town markets.
- Triopoly, quadropoly, and quintopoly coefficients are negative, while monopoly and duopoly prices are similar.
- Hypothesis tests generally do not reject equality of monopoly and duopoly coefficients, nor equality of triopoly through quintopoly coefficients.
- The results support the idea that prices fall when the third firm enters and then level off within the three-to-five firm range.

Detailed interpretation: Table 11 is important because it uses actual prices, not just firm counts. It confirms the entry-threshold interpretation: market conduct changes sharply between one/two firms and three or more firms. However, the urban dummy shows that small-market oligopoly prices remain above large urban-market prices, so the three-to-five firm market is not necessarily fully competitive in an absolute sense.

Economic message: The paper's indirect entry-threshold evidence is consistent with direct price evidence: the second and especially third entrant do most of the work in disciplining prices.

8 Result map

- **Theory:** Entry thresholds summarize the zero-profit market size needed to support each number of firms.
- **Simulation benchmark:** Cournot examples show threshold ratios and margins declining at a decreasing rate as N rises.
- **Sample design:** Isolated local markets create a setting in which local population is a meaningful demand measure.
- **Main estimates:** Threshold ratios fall rapidly; much of the competitive effect occurs by the second or third entrant.

TABLE 11

TIRE PRICE REGRESSIONS (N = 282)

VARIABLE NAME	ORDINARY LEAST SQUARES (1)	ORDINARY LEAST SQUARES (2)	LEAST ABSOLUTE DEVIATIONS (3)
Constant term	26.4	29.9	29.5
	(4.69)	(4.87)	(4.43)
Monopoly market dummy	1.88	.26	.54
	(2.12)	(2.33)	(2.12)
Duopoly market dummy	1.88	−.62	.96
		(2.42)	(2.30)
Triopoly market dummy	−1.80	−2.60	−2.12
	(2.05)	(2.34)	(2.11)
Quadropoly market dummy	−1.80	−3.36	−2.53
		(2.21)	(2.01)
Quintopoly market dummy	−1.80	−1.99	−2.00
		(2.22)	(2.01)
Urban market dummy	−12.1	−11.0	−11.4
	(2.62)	(2.62)	(2.38)
Mileage rating	.43	.38	.39
	(.05)	(.05)	(.05)
County retail wage	1.00	.62	.74
	(.53)	(.53)	(.49)
Other dummy variables	Michelin brand	11 brands	11 brands
Regression R^2	.43	.51	
F or χ^2 hypothesis tests:			
$\alpha_1 = \alpha_2$	.01	.01	1.1
$\alpha_3 = \alpha_4 = \alpha_5$	.68	.70	2.3
$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5$	2.82*	2.86*	448*

NOTE.—The omitted category is all towns not satisfying our monopoly market definition. The numbers in

Table 11: Tire price regressions ($N = 282$)

- **Robustness:** Fixed-cost tests, market-size tests, and profit-shifter tests probe alternative explanations.
- **Market definition:** Stronger isolation criteria yield results consistent with the market-boundary logic.
- **Price validation:** Tire prices confirm that prices fall with early entry and then level off among three-to-five-firm markets.

9 Short summary

- Bresnahan and Reiss ask how actual entry changes competition in concentrated markets.
- They develop entry thresholds, the market size needed to support N firms, as empirical objects.
- Entry threshold ratios provide scale-free measures of the incremental competitive impact of entry.
- The paper studies five local industries in 202 isolated towns.
- The econometric model treats the observed number of firms as an ordered market-structure outcome.
- Most of the competitive change occurs by the entry of the second or third firm.
- Additional entry beyond three to five firms has small incremental effects in the studied markets.
- Tire-price data validate the entry-threshold interpretation.
- The method is especially useful when researchers observe market structure but not prices or costs.

10 Conclusion

10.1 Core conclusion

The paper shows that the competitive consequences of entry can be inferred from market structure data alone. Entry thresholds provide a bridge between theory and empirics: they translate firm counts and market size into information about margins, fixed costs, and conduct.

10.2 Economic intuition

If a monopolist earns large margins, it takes a much larger market to make room for a second firm. If competition is already intense, the next firm requires only roughly proportional increases in market size. Entry-threshold ratios therefore reveal how much the next entrant changes the profitability environment.

10.3 Incremental contribution revisited

The paper's most important contribution is methodological: it shows how to study competition without direct price or cost data. Its second contribution is substantive: in the local retail and professional markets studied, the main competitive action occurs early. The third contribution is empirical validation: tire prices show the same qualitative pattern as the entry-threshold estimates.

10.4 Limitations

The method depends on market definitions, assumptions about cost differences, and the interpretation of cross-sectional entry patterns as equilibrium comparative statics. It does not model dynamic entry timing, overlapping markets, or strategic entry deterrence in detail.

10.5 Takeaway

Bresnahan and Reiss provide a foundational empirical IO tool: use market size and firm counts to infer the competitive effects of entry. The message is not simply that more firms matter; it is that the marginal competitive impact of entry is highly nonlinear and concentrated in the first few entrants.